



Hospital Management System

Objective:



The primary goal of the Hospital Management Service project is to develop a sophisticated and efficient system that facilitates seamless management of various entities within a hospital environment. Implement efficient data handling and storage mechanisms, ensuring seamless interaction with the underlying database to store and retrieve information related to hospital entities.

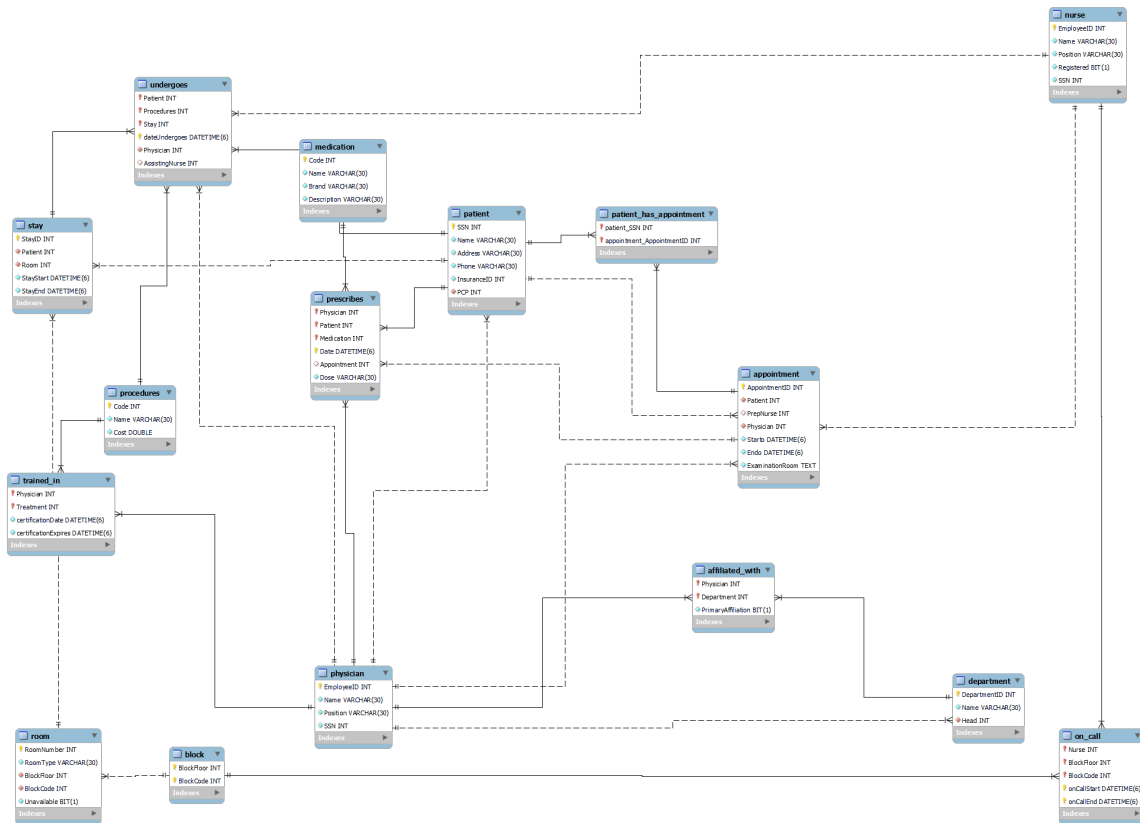
Design Guidelines:

- All Rest API request and response format must be in JSON.
- 200, 201, 400, 404, 401, 500, 505 – use appropriate http status code in your response object.
- You should not alter database schema under any circumstances except wherever changes are mentioned explicitly..
- Use standard HTTP status codes to indicate the success or failure of an API request.

- Adhere to RESTful principles, using HTTP methods (GET, POST, PUT, DELETE) for CRUD operations.

Database Schema

▼ ER Diagram



▼ MySQL Script Files

[create_table.sql](#)

[insert_data.sql](#)

Rest API Design:

All REST API requests and responses follow the JSON format. The project employs standard HTTP status codes (200, 201, 400, 404, 401, 500, 505) appropriately in response objects to convey the outcome of API requests..

HTTP Status Codes

HTTP Status Code	Response
400	Bad Request
401	Unauthorized
403	Forbidden
404	Not Found
500	Internal Server Error
501	Not Implemented

Working with GitHub

▼ Create a New Repository on GitHub

- Log in to your GitHub account.
- Click on the "+" icon in the top right corner and select "New repository."
- Provide a name, description, and other relevant details.

▼ Fork The Repository

- Go to the GitHub page of the repository you want to contribute to.
- Click on the "Fork" button in the upper right corner of the page.

▼ Clone Your Forked Repository:

Clone your forked repository to your local machine.

```
git clone https://github.com/your-username/repository.git
```

▼ Make Changes, Commit, and Push:

- Make the necessary changes to the code.
- Stage and commit your changes.

```
git add .
git commit -m "Your descriptive commit message"
git push
```

▼ Create a Pull Request:

- Go to the GitHub page of your forked repository.
- Switch to the branch you just pushed.
- Click on the "New Pull Request" button.

▼ Review and Merge:

- Describe your changes and submit the pull request.
- The repository owner or maintainers will review your changes.

▼ Update Your Fork (Optional):

To keep your fork up-to-date with the original repository, fetch and merge changes.

```
git fetch upstream
git merge upstream/main
```

Endpoints

Physician

Endpoint	Type	Description	Success Response	Error Response
/api/physician/post	POST	Adds a new Physician	Record Created Successfully	
/api/physician/name/{name}	GET	Search Physician detail by Name	Physician	
/api/physician/position/{pos}	GET	Search Physicians by Position	Collection of Physician	

Endpoint	Type	Description	Success Response	Error Response
/api/physician/empid/{empid}	GET	Search Physician By EmpId	Physician	
/update/position/{position}/{employeeid}	PUT	Update Position Of Physician	Physician	{"timeStamp":"2023-06-08", "message": "Validation failed ..."} ...
/api/physician/update/name/{empid}	PUT	Update Name of an Physician	Physician	
/api/physician/update/ssn/{empid}	PUT	Update SSN of a Physician	Physician	
 /api/department	POST	Add new Department object in DB	Record Created Successfully	
 /api/department/	GET	Get the list of department	Collection of Department	
 /api/department/{deptid}	GET	Search the department detail by dept Id	Department	
/api/department/head/{deptid}	GET	Search the head of department details	Physician	
/api/department/headcertification/{deptid}	GET	Search the head certification detail by deptid	Collection of Certification	
/api/department/{head}	GET	Search the list of department by head id (physician id)	Collection of Department	
/api/department/check/{physicianid}	GET	Search whether a physician is head of any department or not	Boolean	
/api/department	PUT	Update the department head id	Department	

Endpoint	Type	Description	Success Response	Error Response
<u>/update/headid/{deptid}</u>				
<u>/api/department/update/deptname/{deptid}</u>	PUT	Update the name of department	Department	
 <u>/api/affiliated_with/post</u>	POST	Add a existing physician to affiliated_with Table	String: Record Created Successfully	
<u>/api/affiliated_with/physicians/{deptid}</u>	GET	Search a list of physician affiliated with particular department	Collection of Physician	
<u>/api/affiliated_with/department/{physicianid}</u>	GET	Search the list of department for a physician	Collection of Department	
<u>/api/affiliated_with/countphysician/{deptid}</u>	GET	Count total number of physician in a department	int	
<u>/api/affiliated_with/primary/{physicianid}</u>	GET	Search the primary affiliation of physician	boolean	
<u>/api/affiliated_with/primary/{physicianid}</u>	GET	Update the primary affiliation of physician	boolean	
<u>/api/procedure</u>	POST	Add a new treatment (procedure) to DB	String: Record Created Successfully	
<u>/api/procedure/</u>	GET	Get a list of available Treatment	List of Procedure	

Endpoint	Type	Description	Success Response	Error Response
/api/procedure/cost/{id}	GET	Get the cost of procedure by id	Procedure	
/api/procedure/cost/{name}	GET	Search the cost of procedure by name	Procedure	
/api/procedure/cost/id	PUT	Update the cost of procedure	Procedure	
/api/procedure/name/{id}	PUT	Update the name of Procedure	Procedure	
/api/trained_in	POST	Add a certification to a DB	String: Record Created Successfully	
/api/trained_in/	GET	Get a list of Procedure available with certification	Collection of Procedure	
/api/trained_in/treatment/{physicianid}	GET	Get a list of treatment can be done by particular physician	Collection of Procedure	
/api/trained_in/physicians/{procedureid}	GET	Get a list of physician for a particular treatment	Collection of Physician	
/api/trained_in/expiredsooner ti/{physicianid}	GET	Get a list of Procedure whose certification expire within a month	Collection of Procedure	
/api/trained_in/certificationexpiry/{physicianid&procedureid}	PUT	Update the certification expiry date in the basis if physician id and procedureid	boolean	
api/patient	POST	Add a patient report to the DB	String: Record Created Successfully	
api/patient/	GET	Get a list of Patient	Collection of Patient	

Endpoint	Type	Description	Success Response	Error Response
api/patient/{physicianid}	GET	Get a list of patient who is check by a particular physicians	Collection of Patient	
api/patient/{physicianid}/{patientid}	GET	Get detail of patient checked by a particular physician	Patient	
api/patient/insurance/{patientid}	GET	Get the insuranceid of patient	Int	
api/patient/address/{SSN}	PUT	Update the address of a patient by using SSN	Patient	
api/patient/phone/{SSN}	PUT	Update the phone of a patient by using SSN	Patient	
api/nurse	POST	Add a nurse detail to DB	String: Record Created Successfully	
api/nurse/	GET	Get a list of nurse	Collection of nurse object	
api/nurse/{empid}	GET	Get a detail of nurse by employeeid	Nurse	
api/nurse/position/{empid}	GET	Get position of nurse by employeeid	String	
api/nurse/registered/{empid}	GET	Get about a nurse is registered or not	boolean	
api/nurse/registered/{empid}	PUT	Update the value of registered by employeeid	Nurse	
api/nurse/ssn/{empid}	PUT	Update the value of SSN by empid	Nurse	
api/appointment/	POST	Add a appointment detail to DB	String: Record Created Successfully	

Endpoint	Type	Description	Success Response	Error Response
api/appointment	GET	Get a list of appointment	Collection of Appointment	
api/appointment/{startdate}	GET	Get a list of appointment by startdate	Collection of appointment	
api/appointment/patient/{appointmentid}	GET	Search a patient information by appointment id	Patient	
api/appointment/physician/{appointmentid}	GET	Search a physician detail by appointment id	Physician	
api/appointment/nurse/{appointmentid}	GET	Search a nurse detail by appointment id	Nurse	
api/appointment/examinationroom/{appointmentid}	GET	Search a examination room detail by appointment id	Room	
api/appointment/physician/{patientid}	GET	Get a list of physician by patient id	Collection of Physician	
api/appointment/physician/{patientid}/{date}	GET	Get a physician detail by patient id on a particular date	Physician	
api/appointment/nurse/{patientid}	GET	Get a list of Nurse by patient id	Collection of Nurse	
api/appointment/nurse/{patientid}/{date}	GET	Get a physician detail by Nurse id on a particular date	Nurse	

Endpoint	Type	Description	Success Response	Error Response
api/appointments/date/{patientid}	GET	Get a list of dates on which a patient has an appointment	Collection of Date	
api/appointments/patient/{physicianid}	GET	Get a list of patient checked by physician	Collection of Patient	
api/appointments/patient/{physicianid}/{date}	GET	Get a list of patient checked by physician on particular date	Collection of Patient	
api/appointments/patient/{physicianid}/{patientid}	GET	Get about a patient checked by physician by patient id	Patient	
api/appointments/patient/{nurseid}	GET	Get a list of patient checked by nurse	Collection of Patient	
api/appointments/patient/{nurseid}/{patientid}	GET	Get about a patient checked by nurse by patient id	Patient	
api/appointments/patient/{nurseid}/{date}	GET	Get a list of patient checked by Nurse on particular date	Collection of Patient	
api/appointments/room/{patientid}/{date}	GET	Get room details by patient id on a particular date	Room	
api/appointments/room/{physicianid}/{date}	GET	Get a list of room details by physician id on a particular date	Collection of Room	
api/appointments/room/{nurseid}/{date}	GET	Get a list of room details by Nurse id on a particular date	Collection of Room	
api/appointments/room/{appointmentid}	PUT	Update a examination room by appointment id	Room	

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